Run Ad-Hoc Copy and Run (adhocr)

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Who is Gratien D'haese?

- ***Independent UNIX Consultant***
- ***Over 25 years of experience with UNIX (using Linux since Dec 1991 version 0.1)***
- ***…source projects involved:***

  - ***Relax-and-Recover***
  - ***Make CD-ROM Recovery (dev on hold)***
  - ***WBEMextras (towards HP-UX HPSIM clients)***
  - ***Ad-hoc Copy and Run (adhocr)***

So What ?

- ***ADHOCR stands for Ad-Hoc Copy and Run commands on remote Unix systems***
- ***Nice – SSH and/or SCP do the same, right?***
- ***However, in some organisations it is not that simple to use ssh & scp as “root”***
- ***Fine – SUDO is the answer***
- ***Yes, however, in some organisations***

Confused?

- ***Indeed, sometimes it gets the form of a real bureaucracy to get something done***
- ***Security, logging, evidence, segregation of duties make our lives as system administrators not easy***
- ***The opposite of bureaucracy is adhocracy – be flexible and responsive to the needs of the moment***
- ***Bonsai: strip 'till the essentials***

Challenges

- ***Amount of systems in global organisations***
  - ***Old systems get decommissioned***
  - ***New systems are set-up***
  - ***In a global organisation no-one really knows how many systems disappear or being added (monthly extract from central management database)***
  - ***On most systems Secure Shell keys were exchanged, but we lost track of it***

What can adhocr do for you?

- ***Run commands on remote Unix systems (Linux, HP-UX, Solaris, AIX, ...)***
  - ***Under your account***
  - ***As 'root' via 'sudo su -'***
- ***Enter your password only once***
  - ***Ideal in Active Directory environments, LDAP integration with e.g. centrify***
  - ***“sudo su –“ must be execute under your account***

What can adhocr do for you?

- ***Central point of logging***
- ***Output of running commands collected in <u>one</u> output file (or optional per system)***
- ***Batch mode***
- ***Parallellization***
- ***Easy error reporting (at the end of the batch)***

Adhocr building blocks

- *Written in Korn shell (or Bash)*
- *Secure Shell*
- *Requires expect tool:*
    - *Programmed dialogue with interactive programs, e.g. telnet, ftp, ssh, sftp, etc...*
    - *Written by Don Libes between 1987 and 1999*
    - *Home page: http://expect.nist.gov*
    - *Learning expect – see README of expect*

Re-inventing the wheel?

- ***Probably 'adhocr' seems nothing new?***

"I thought I was on to something but I can't figure out how to move it."

Inventing the Wheel cartoon, October 2, 2009.
(Bill Abbott http://www.toonpool.com/)

Alternatives (1)

- ***Parallel-ssh - http://code.google.com/p/parallel-ssh/***
- ***Enhanced parallel-ssh with modules and scripts https://github.com/jcmcken/parallel-ssh***
- ***pssh -h hostfile.txt --script restart_iptables.sh –sudo***
- ***Still expecting sudo without password prompting***
- ***Written in python***

Alternatives (2)

- ***Parallel Distributed Shell - https://code.google.com/p/pdsh/***
- ***`pdsh -R ssh -w host1,host2 command`***
- ***Expects ssh keys have been exchanged***
- ***Sudo is not native foreseen***
- ***Written in C language***

Alternatives (3)

- ***Fabric - https://github.com/fabric/fabric***
- ***Python library and command-line tool for streamlining the use of SSH for application deployment or systems administration tasks***
- ***Seems to be python version dependent***
- ***Problematic to use on different UNIXes***

Alternatives (4)

- ***Rex - http://rexify.org/***
- ***Manage from a central point through the complete process of configuration management and software deployment***
- `rex -e 'say run "uptime";' -H "hosts[01..10]" -u root -p password`
- ***Written in perl***
- ***Complicated tasks need rexfiles***
- **Requires a learning curve**

Alternatives (5)

- ***Func (Fedora Unified Network Controller) - https://fedorahosted.org/func/***
- ***Written in python and needs certmaster***
- ***Is Linux focused***
- *`func *.domain.com call hardware info`*
- ***Not really an option in our organisation***
- ***Not too complicated if used as SSH***

Alternatives (6)

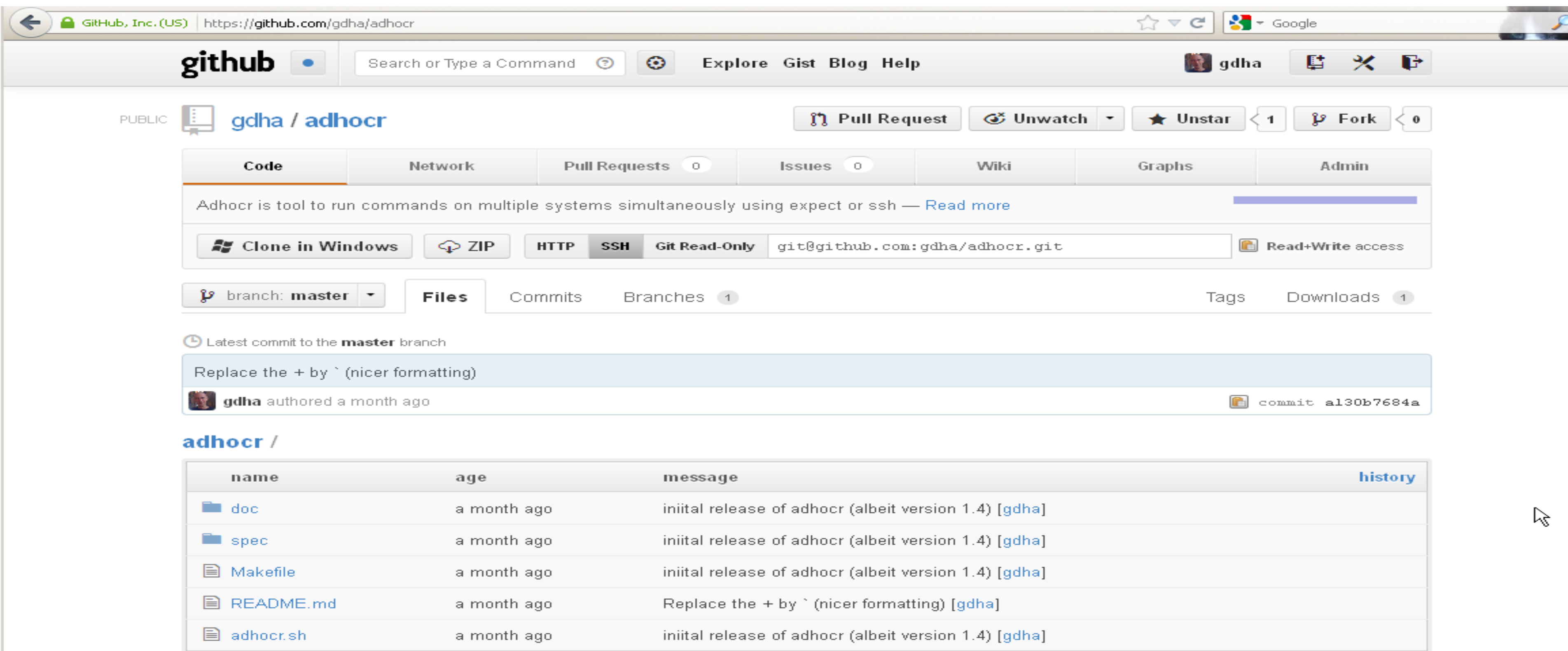
- ***Ansible - http://ansible.github.com/***
- ***Written in python***
- ***Uses SSH and has no other dependencies***
- ***Ansible has a short learning curve***
- ***`ansible atlanta -a "commands" -u username --sudo [--ask-sudo-pass]`***
- ***Comes very close to what we need***
- ***Sudo to root (without password prompting)***

Tips and Tricks (1)

- ***Distributing your public key***
  - ***ssh-copy-id -i ~/.ssh/id_rsa.pub user @server***
  - ***Pitty ssh-copy-id command is not available on all Unix versions.***
- ***Distributing your public key (alternative)***
  - ***Use adhocr for this task***
- ***Play with Ansible playbooks (very attractive)***

Adhocr home page

- ***https://github.com/gdha/adhocr***
- ***git clone git@github.com***



The expect magic

```
VAR=$(expect -c "
set password \$env("PASS") ;
spawn ssh $SSHoptions $USER@$HOST  $CMD
match_max 100000 ;
set timeout 10 ;
expect  {
        \"(yes/no)?\" { send -- \"yes\\r\" } ;
        \"*?assword:*\" {
                send -- \"\$password\\r\" ;
                expect -re \"\[\$@#>] $\" ;
                }
}

 wait
")      # end-of-expect VAR

echo "$VAR" >$LOGDIR/$DATE_TIME/$2 2>&1
```

Makefile (Linux only)

```
rewrite:
	@echo -e "\033[1m== Rewriting $(adhocr_source) ==\033[0;0m"
	sed -i.orig \
		-e 's#^Version=.*#Version=$(version)#' \
		-e 's#^CompanyName=.*#CompanyName=$(companyname)#' \
		-e 's#^SudoGroup=.*#SudoGroup=$(sudogroup)#' \
		$(adhocr_source)

adhocr: adhocr.sh.x
	-cp -f adhocr.sh.x adhocr
	-chmod 711 adhocr

adhocr.sh.x: $(adhocr_source) rewrite shc
	/usr/local/bin/shc -r -T -f $(adhocr_source)

shc:
	@echo -e "\033[1m== Shell Compiling $(adhocr_source) ==\033[0;0m"
	if test ! -x $(shc_bin) ; then \
		@echo "Error: we need shc (http://www.datsi.fi.upm.es/~frosal/)" ; \
		exit 1 ; \
	fi
```

Tips and Tricks (2)

- ***Shell Compiling : Source code protection***
  - ***Tired of customers using your trial scripts (free development)?***
- ***Try SHC from Francisco Rosales***
  - ***Encrypts the shell script, and puts a C wrapper around it***
  - ***http://www.datsi.fi.upm.es/~frosal/***
  - ***Does not compile on all OSes***
  - ***Remember security by obscurity is no good***

adhocr.spec file

```
$ more spec/adhocr.spec
%define rpmrelease %{nil}
%define companyname "Your Company Name"
%define sudogroup "wheel"

Summary: A tool to run commands on multiple systems
    simultaneously using expect
Name: adhocr
Version: 1.4
Release: 1%{?rpmrelease}%{?dist}
License: GPLv3
Group: Applications/File
URL: https://github.com/gdha/adhocr
```

Installation of rpm (Linux)

- *$ make rpm*
- *$ sudo rpm -ivh adhocr-1.4-1.el6.x86_64.rpm*
  error: Failed dependencies:
  expect is needed by adhocr-1.4-1.el6.x86_64
  ksh is needed by adhocr-1.4-1.el6.x86_64
- *Install the missing dependencies*
- *$ file /usr/bin/adhocr*
  /usr/bin/adhocr: ELF 64-bit LSB
  executable, x86-64, version 1 (SYSV)

Installation on non-Linux

- ***The adhocr.sh is the only script that is needed***
- ***Customise 2 parameters:***
  - ***CompanyName***
  - ***SudoGroup***
- ***Copy script to /usr/local/bin/adhocr***
- ***Keep in mind the dependencies for***
  - ***Ksh***
  - ***Expect***

Tips and Tricks (3)

- ***To install dependencies on HP-UX use depothelper (free)***
  - ***http://hpux.connect.org.uk/hppd/hpux/Sysadmin/ \ depothelper-2.00/***
  - ***# bin/depothelper expect***
- ***On Windows use Cygwin (free)***
  - ***Run setup.exe and select ksh and expect***
- ***Solaris: https://unixpackages.com/ (not free)***

Adhocr usage

$ adhocr

```
************************************************
     adhocr : Ad-hoc Copy and Run
           version 1.4
************************************************

Usage: adhocr [-p #max-processes] [-u username] [-k] -f filename-containing-systems [-h] -c "commands to execute"
        -p maximum number of concurrent processes running (in the background) [optional - default is 10]
        -u The user "username" should be part of the "se" group for executing sudo [default is gdha]
        -k keep the log directory with individual log files per system [optional - default is remove]
        -f filename containing list of systems to process
        -h show extended usage
        -c "command(s) to execute on remote systems"
```

Extended help (1)

- *-p # threads (Maximum number of concurrent processes running)*
- *-u <username> (by default your account)*
- *-k (keep the log directory)*
- *-f <filename> (containing list of systems)*
- *-l <logdir> (by default . or logs/ if it exists)*
- *-o <outputdir> (by default . or*

Extended help (2)

- ***-x (use expect – is default behaviour)***
- ***- npw|-nx|-bg (use only SSH keys) !***
- ***-up (upload files)***
- ***-dl (download files)***
- ***-t <timeout> (in secs to kill hanging procs)***
- ***-h show extended help***
- ***-c <command(s)>***

Simple queries

$ adhocr -f HPUX1111-systems -t 30 -p 50 -c uptime

```
************************************************
      adhocr : Ad-hoc Copy and Run
              version 1.4
************************************************

  ** Enter the domain password of user gdhaese:
Script name : /usr/bin/adhocr
Filename containing list of systems : HPUX1111-systems
Amount of systems to roll-over is 334
Will execute the commands in a bunch of 50
Command to execute : uptime
The individual log files found under ./2012-10-19.153459 will be removed at the end

[1] Executing expect with ssh gdhaese1@brsjd002 uptime
======= brsjd002 (starting at 101912_1535)
```

Run adhocr as another user (1)

**# adhocr -u gdhaese -f systems/tape-hosts -t 30 **
-c /home/gdhaese/bin/check_san_tape_device.sh

```
**************************************************
     adhocr : Ad-hoc Copy and Run
           version 1.4
**************************************************

 ** Enter the domain password of user gdhaese:
Script name : adhocr
Filename containing list of systems : systems/tape-hosts
Amount of systems to roll-over is 2
Will execute the commands in a bunch of 10
Command to execute : /home/gdhaese/bin/check_san_tape_device.sh
The individual log files found under ./logs/2012-10-18.160819
will be removed at the end
.....
```

Run adhocr as another user (2)

```
...
[1] Executing expect with ssh gdhaese@mdde1d01  \
/home/gdhaese/bin/check_san_tape_device.sh
======= mdde1d01 (starting at 101812_1608)
[2] Executing expect with ssh gdhaese@mdde1d02  \
/home/gdhaese/bin/check_san_tape_device.sh
======= mdde1d02 (starting at 101812_1608)
      - 2 running jobs at this moment.
======= mdde1d01 (ending at 101812_1608)
======= mdde1d02 (ending at 101812_1608)

*** Logfile = ./logs/adhocr-2012-10-18.160819.log
     (containing error messages)
*** Output  = ./output/adhocr-2012-10-18.160819.output
     (concatenated output of system output)
..
*** Removing Output directory ./logs/2012-10-18.160819/
```

Security considerations

- ***gdha 15982 15973 0 16:55 pts/0 00:00:00 expect -c ?set password $env(PASS) ; ?spawn ssh -o ConnectTimeout=10 -o StrictHostKeyChecking=no gdhaese@itsusmlfean08 rpm -q rear ? match_max 100000 ; ?set timeout 10 ; ?expect { ??"(yes/no)?" { send -- "yes\r" } ; ??"*?assword:*" { ???send -- "$password\r" ; ???expect -re "\[$@#>] $" ; ???} ?} ?# send -- "\r" ; ?# expect -re "\[$@#>] $" ; ?# send -- "rpm -q***

Uploading files with adhocr

- ***To upload scripts or other files to selected hosts use***
- ***adhocr -f systems -t 30 -up -c “local-file remote-location”***
- ***adhocr -f systems -c "mkdir -m 700 .ssh"***
- ***adhocr -t 60 -f systems -up -c "~/.ssh/authorized_keys .ssh/"***

Executing tasks with adhocr

- ***adhocr -f systems -t 30 -up -c "adhocr_rear_upgrade.sh bin/"***
- ***adhocr -f systems -t 30 -c "/home/gdha/bin/adhocr_rear_upgrade.sh" -sudo***

```
************************************************
       adhocr : Ad-hoc Copy and Run
                version 1.4
************************************************


###################################################################
                          S U D O    W A R N I N G
###################################################################
 You are about to be granted root shell access. By continuing,
 you agree to the following requirements:
 ....
```

Output cluttered with sudo stuff

- ***The output file is not really readable with all the sudo output***

```
BEGIN HOST ##### itsusralabvm029 #####
spawn ssh -o ConnectTimeout=10 -o StrictHostKeyChecking=no gdhaese@itsusralabvm029
gdhaese@itsusralabvm029's password:
Last login: Thu Oct 25 04:30:08 2012 from itsusralabvm029
gdhaese@itsusralabvm029:~>
gdhaese@itsusralabvm029:~> sudo su -

 You are about to be granted root shell access. By continuing, you agree to
 the following requirements:

   - Your access to the root shell must have been authorized by being a member
     of one of the groups that grants this access.
   - You may not use the privileges granted by the use of the root shell to
     grant elevated privileges to any other user or any other account.
   - If you have been granted root shell access on a temporary basis, you MUST
     exit the root shell as soon as you complete your actions.

 Unauthorized use may subject you to My Company disciplinary proceedings
 and/or criminal and civil penalties under state, federal or other applicable
 domestic and foreign laws. The use of this system may be monitored and recorded
 for administrative and security reasons. If such monitoring and/or recording
 reveal possible evidence of criminal activity, My Company may provide
 the evidence of such monitoring to law enforcement officials.

gdhaese's password:
[root@itsusralabvm029:/root]#
#->
[root@itsusralabvm029:/root]#
#-> /home/gdhaese1/adhocr_rear_upgrade.sh
--------------------------------------------------------------------------------
              Script: adhocr_rear_upgrade.sh
    Installation Host: itsusralabvm029
    Installation User: root
    Installation Date: Thu Oct 25 08:35:46 UTC 2012
     Installation Log: /var/adm/install-logs/adhocr_rear_upgrade.scriptlog
--------------------------------------------------------------------------------

 *** Pre-installation Test on system itsusralabvm029.dfdev.jnj.com ***
rear-1.14-3
```

Using start-end markers

- **#=-=-=# Start ... #=-=-=# End**

```
cat ./adhocr-2012-10-25.071012.output
BEGIN HOST ##### itsusralabvm029 #####
-------------------------------------------------------------------------------
               Script: adhocr_rear_upgrade.sh
    Installation Host: itsusralabvm029
    Installation User: root
    Installation Date: Thu Oct 25 11:10:28 UTC 2012
     Installation Log: /var/adm/install-logs/adhocr_rear_upgrade.scriptlog
-------------------------------------------------------------------------------

 *** Pre-installation Test on system itsusralabvm029 ***
rear-1.14-3
-------------------------------------------------------------------------------
 *** Installation Steps on system itsusralabvm029 ***
Upgrading rear
Loading repository data...
Reading installed packages...
'rear' is already installed.
Resolving package dependencies...

Nothing to do.
-------------------------------------------------------------------------------
 *** Post-installation Test on system itsusralabvm029 ***
rear-1.14-3
-------------------------------------------------------------------------------


-------------------------------------------------------------------------------
 ** Script ended at Thu Oct 25 11:10:30 UTC 2012
Execution time on host itsusralabvm029 was 11.4928730220794678 seconds
END HOST ##### itsusralabvm029 #####
-------------------------------------------------------------------------------
```

adhocr or not to adhocr?

- ***Run commands on remote Unix systems (Linux, HP-UX, Solaris, AIX, ...)***
  - ***Under your account***
  - ***As 'root' via 'sudo su -'***
- ***Enter your password only once***
  - ***Ideal in Active Directory environments, LDAP integration with e.g. centrify***
  - ***“sudo su –“ must be execute under your account***

Demo time & QA